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United States Patent

12410725

Kind Code

B1

Date of Patent

September 09, 2025

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Turbine shroud assemblies with air activated pistons for biasing buffer cavity seals

Abstract

A turbine shroud assembly for use with a gas turbine engine includes a carrier segment, a blade track segment, and a seal system. The carrier segment arranged circumferentially at least partway around an axis. The blade track segment is coupled to the carrier segment and defines a portion of a gas path of the gas turbine engine. The seal system is arranged radially between the carrier segment and the blade track segment to block gases from flowing between the carrier segment and the blade track segment.

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Family ID: 96950391

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Appl. No.: 18/680758

Filed: May 31, 2024

Publication Classification

Int. Cl.: F01D11/08 (20060101)

U.S. Cl.:

CPC F01D11/08 (20130101); F05D2230/60 (20130101); F05D2240/11 (20130101); F05D2240/55 (20130101); F05D2260/30 (20130101)

Field of Classification Search

CPC: F01D (11/04); F01D (11/005); F01D (11/08); F01D (11/10); F01D (11/14); F02C (7/28); F05D (2240/11); F05D (2240/55); F05D (2260/30); F05D (2230/60)

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to turbine shroud assemblies, and more specifically to sealing of turbine shroud assemblies used with gas turbine engines.

BACKGROUND

(2) Gas turbine engines are used to power aircraft, watercraft, power generators, and the like. Gas turbine engines typically include a compressor, a combustor, and a turbine. The compressor compresses air drawn into the engine and delivers high pressure air to the combustor. In the combustor, fuel is mixed with the high pressure air and is ignited. Products of the combustion reaction in the combustor are directed into the turbine where work is extracted to drive the compressor and, sometimes, an output shaft. Left-over products of the combustion are exhausted out of the turbine and may provide thrust in some applications.

(3) Compressors and turbines typically include alternating stages of static vane assemblies and

rotating wheel assemblies. The rotating wheel assemblies include disks carrying blades around their outer edges. When the rotating wheel assemblies turn, tips of the blades move along blade tracks included in static shrouds that are arranged around the rotating wheel assemblies. Such static shrouds may be coupled to an engine case that surrounds the compressor, the combustor, and the turbine.

(4) Some shrouds positioned in the turbine may be exposed to high temperatures from products of the combustion reaction in the combustor. Such shrouds sometimes include components made from materials that have different coefficients of thermal expansion. Due to the differing coefficients of thermal expansion, the components of some turbine shrouds expand at different rates when exposed to combustion products. In some examples, sealing between and coupling such components may present challenges.

SUMMARY

(5) The present disclosure may comprise one or more of the following features and combinations thereof.

(6) A turbine shroud assembly adapted for use with a gas turbine engine may include a carrier segment, a blade track segment, and a seal system. The carrier segment may be arranged circumferentially at least partway around an axis. The blade track segment may be arranged circumferentially at least partway around the axis to define a portion of a gas path of the gas turbine engine. The seal system may be arranged radially between the carrier segment and the blade track segment to block gases from flowing between the carrier segment and the blade track segment.

(7) In some embodiments, the carrier segment may include an outer wall, a first support wall, and a second support wall. The first support wall may extend radially inward from the outer wall. The second support wall may extend radially inward from the outer wall at a location spaced apart axially from the first support wall to define an attachment-receiving space. The first support wall may be formed to include a radially-inwardly opening first channel that extends circumferentially relative to the axis and at least one buffer air passageway that extends radially into the forward support wall and opens into the first channel.

(8) In some embodiments, the blade track segment may include a shroud wall and an attachment feature. The shroud wall may extend circumferentially partway around the axis. The attachment feature may extend radially outward from the shroud wall into the attachment-receiving space formed in the carrier segment.

(9) In some embodiments, the seal system may include a buffer air seal assembly arranged in the first channel between the forward support wall of the carrier segment and the shroud wall of the blade track segment to block gases from flowing between the carrier segment and the blade track segment. The buffer air seal assembly may include a seal. The seal may extend circumferentially relative to the axis and axially across the first channel.

(10) In some embodiments, the at least one buffer air passageway may be configured to discharge buffer air radially inward away from the carrier segment into the first channel. The at least one buffer air passageway may be configured to discharge buffer air radially inward away from the carrier segment into the first channel to urge the seal radially inward into engagement with the shroud wall of the blade track segment.

(11) In some embodiments, the buffer air seal assembly may further include a bias member. The bias member may be compressed radially between the carrier segment and the seal to apply a bias force to the seal to bias the seal radially inward towards the shroud wall of the blade track segment.

(12) In some embodiments, the seal may be formed to include an inner groove and at least one through hole. The inner groove may extend radially inward into the seal and circumferentially relative to the axis. The at least one through hole may extend radially through the seal and opens into the inner groove to allow the buffer air to flow through the seal into the inner groove.

(13) In some embodiments, the seal may include a first seal member and a second seal member. The second seal member may be coupled to the first seal member for movement therewith.

(14) In some embodiments, the first seal member may have a rectangular cross-sectional shape when viewed in the circumferential direction. The rectangular cross-sectional shape of the first seal member may define an inner surface engaged with the shroud wall of the blade track segment, an outer surface spaced apart radially from the inner surface, and two side surfaces spaced apart axially that extend radially between the inner surface and the outer surface. The second seal member may be arranged axially between a first side surface of the first seal member and the carrier segment.

(15) In some embodiments, the first seal member may be formed to include a side groove. The side groove may extend axially into a first side surface of the first seal member and circumferentially relative to the axis. The second seal member may be arranged in the side groove and engage the carrier segment in the first channel.

(16) In some embodiments, the first seal member may be formed to include an inner groove and at least one through hole. The inner groove may extend radially into the inner surface of the first seal member and circumferentially relative to the axis. The at least one through hole may extend radially through the first seal member and opens into the inner groove to allow the buffer air to flow through the first seal member into the inner groove.

(17) In some embodiments, the seal may include an outer seal member, a forward seal member, and an aft seal member. The forward seal member may extend radially inward from the outer seal member, and an aft seal member that extends radially inward from the outer seal member, the aft seal member spaced apart axially from the forward seal member to define an inner groove therebetween, and the outer seal member, the forward seal member, and aft seal member are free to move relative to each other.

(18) In some embodiments, the outer seal member may be formed to include at least one through hole that extends radially through the outer seal member and opens into the inner groove. The outer seal member may be formed to include at least one through hole that extends radially through the outer seal member and opens into the inner groove to allow the buffer air to flow through the outer seal member into the inner groove to urge the forward seal member axially forward and the aft seal member axially aft into engagement with the carrier segment.

(19) In some embodiments, the seal may include a first seal member and a second seal member coupled to the first seal member for movement therewith. The first seal member may have a rectangular cross-sectional shape when viewed in the circumferential direction that defines an inner surface, an outer surface spaced apart radially from the inner surface, and two side surfaces spaced apart axially that extend radially between the inner surface and the outer surface. The second seal member may extend around a portion of the first seal member so that the second seal member engages the carrier segment and the blade track segment.

(20) In some embodiments, the first seal member may be formed to include an inner groove and at least one through hole. The inner groove may extend radially into the inner surface of the first seal member and circumferentially relative to the axis. The at least one through hole may extend radially through the first seal member and opens into the inner groove to allow the buffer air to flow through the first seal member into the inner groove. In some embodiments, the first channel may have a rectangular cross-sectional shape viewed circumferentially relative to the axis.

(21) In some embodiments, the first channel may define a first end surface, a first support-wall surface, and a second support-wall surface. The first support-wall surface may extend radially inward from the first end surface. The second support-wall surface may extend radially inward from the first end surface at a location spaced apart axially from the first support-wall surface.

(22) According to another aspect of the present disclosure, a method may include providing a carrier segment arranged circumferentially at least partway around an axis. The carrier segment may be formed to include a radially-inwardly opening first channel and at least one buffer air passageway that extends radially into the carrier segment.

(23) In some embodiments, the method may further include providing a blade track segment

arranged circumferentially at least partway around the axis. The blade track segment may have a shroud wall that extends circumferentially partway around the axis and an attachment feature that extends radially outward from the shroud wall. The shroud wall may be formed to include a radially-outwardly opening second channel.

(24) In some embodiments, the method may further include providing a buffer air seal assembly. The buffer air seal assembly may include a seal that extends circumferentially relative to the axis and at least one bias member.

(25) In some embodiments, the method may further include arranging the at least one bias member of the buffer air seal assembly in the first channel formed in the carrier segment, arranging the seal of the buffer air seal assembly in the first channel formed in the carrier segment so that the bias member is located radially between the carrier segment and the seal, and arranging the blade track segment adjacent to the carrier segment so that the seal engages the shroud wall of the blade track segment. In some embodiments, the method may further include discharging a flow of buffer air through the at least one buffer air passageway radially inward away from the carrier segment toward the seal to urge the seal radially inward into engagement with the blade track segment.

(26) In some embodiments, the seal may include a first seal member and a second seal member. In some embodiments, arranging the seal of the buffer air seal assembly in the first channel formed in the carrier segment may include arranging the second seal member in a side groove formed in the first seal member and arranging the first and second seal members in the first channel formed in the carrier segment after arranging the second seal member in the side groove of the first seal member.

(27) In some embodiments, the seal may include an outer seal member, a forward seal member, and an aft seal member. In some embodiments, arranging the seal of the buffer air seal assembly in the first channel formed in the carrier segment may include arranging the outer seal member in the first channel so that the at least one bias member is located radially between the carrier segment and the outer seal member, arranging the forward and aft seal members in the first channel after arranging the outer seal member in the first channel, and spacing the aft seal member apart axially from the forward seal member to define an inner groove therebetween.

(28) In some embodiments, the seal includes a first seal member and a second seal member coupled to the first seal member for movement therewith. The first seal member may have a rectangular cross-sectional shape when viewed in the circumferential direction that defines an inner surface, an outer surface spaced apart radially from the inner surface, and two side surfaces spaced apart axially that extend radially between the inner surface and the outer surface. The second seal member may extend around the inner surface of the first seal member.

(29) In some embodiments, the method may further include providing at least one retainer and inserting the at least one retainer axially into the carrier segment and through the attachment feature of the blade track segment. The method may include inserting the at least one retainer axially into the carrier segment and through the attachment feature of the blade track segment to couple the blade track segment to the carrier segment.

(30) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cut-away perspective view of a gas turbine engine showing that the exemplary engine includes a fan, a compressor, a combustor, and a turbine and suggesting that the turbine includes turbine wheel assemblies and static vane assemblies surrounded by a turbine shroud assembly;
- (2) FIG. 2 is a partial cross-sectional view of the gas turbine engine of FIG. 1 showing a portion of

the turbine in which the turbine shroud assembly is located radially outward from blades of a turbine wheel assembly to block gasses from passing over the blades without interacting with the blades, and further showing the turbine shroud assembly includes a carrier segment, a blade track segment coupled to the carrier segment to define a portion of a gas path of the gas turbine engine, and a seal system configured to seal between the carrier segment and the blade track segment to block gases flowing through the gas path from flowing between the carrier segment and the blade track segment;

(3) FIG. 3 is a perspective view of a portion of the turbine shroud assembly of FIG. 2 showing the turbine shroud assembly includes the carrier segment and the blade track segment made from ceramic matrix composite materials coupled to the carrier segment;

(4) FIG. 4 is an exploded view of the turbine shroud assembly of FIG. 3 showing the seal system includes a forward seal assembly having a seal and bias members configured to be arranged in a first channel formed in the carrier segment and an aft seal assembly having a plurality of seal members configured to be arranged in a second channel formed in the carrier segment;

(5) FIG. 5 is a cross-section view of the turbine shroud assembly of FIG. 3 taken along line 5-5 showing the forward support wall of the carrier segment is formed to include the radially-inwardly opening first channel that receives the seal and the bias members of the forward seal assembly and a buffer air passageway configured to discharge buffer air into the first channel to urge the seal radially inward into engagement with the blade track segment, and further showing the shroud wall of the blade track segment is formed to include a radially-outwardly opening second channel that receives the seal;

(6) FIG. 6 is a detail view of FIG. 5 showing the forward seal assembly—also referred to as the buffer air seal assembly—includes the seal having a first seal member that extends radially into the first channel formed in the carrier segment and the second channel formed in the blade track segment and a second seal member arranged in a side groove in the first seal member so that the second seal member is axially between the first seal member and the carrier segment;

(7) FIG. 7 is an exploded view of another turbine shroud assembly included in the gas turbine engine of FIG. 1 showing the turbine shroud assembly includes a carrier segment, a blade track segment, and a seal system having a forward seal assembly configured to be arranged in a first channel formed in the carrier segment and an aft seal assembly configured to be arranged in a second channel formed in the carrier segment, and further showing the forward seal assembly includes a seal made up of a plurality seal members configured to be assembled together in the first channel;

(8) FIG. 8 is a cross-sectional view of the turbine shroud assembly of FIG. 7 showing the forward support wall of the carrier segment is formed to include the radially-inwardly opening first channel that receives the seal of the forward seal assembly and a buffer air passageway configured to discharge buffer air into the first channel to urge the seal radially inward into engagement with the blade track segment; and

(9) FIG. 9 is a detail view of FIG. 8 showing the forward seal assembly—also referred to as the buffer air seal assembly—includes the seal having an outer seal member, a forward seal member that extends radially inward from the outer seal member, and an aft seal member that extends radially inward from the outer seal member and spaced apart axially from the forward seal member to define a buffer air cavity therebetween, and further showing the outer seal member is formed to include at least one through hole that extends radially through the outer seal member to allow the buffer air to flow through the outer seal member into the buffer air cavity to urge the forward seal member axially forward and the aft seal member axially aft into engagement with the carrier segment;

(10) FIG. 10 is an exploded view of another turbine shroud assembly included in the gas turbine engine of FIG. 1 showing the turbine shroud assembly includes a carrier segment, a blade track segment, and a seal system having a forward seal assembly configured to be arranged in a first

channel formed in the carrier segment and an aft seal assembly configured to be arranged in a second channel formed in the carrier segment, and further showing the forward seal assembly includes a seal made up two seal members configured to assembled together in the first channel; (11) FIG. **11** is a cross-sectional view of the turbine shroud assembly of FIG. **10** showing the forward support wall of the carrier segment is formed to include the radially-inwardly opening first channel that receives the seal of the forward seal assembly and a buffer air passageway configured to discharge buffer air into the first channel to urge the seal radially inward into engagement with the blade track segment; and (12) FIG. **12** is a detail view of FIG. **11** showing the forward seal assembly—also referred to as the buffer air seal assembly—includes the seal having a first seal member and a second U-shaped seal member that extends around the first seal member.

DETAILED DESCRIPTION OF THE DRAWINGS

(13) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

(14) A turbine shroud segment **22** is shown in FIGS. **2-6** and is adapted for use in a gas turbine engine **10** as shown in FIG. **1**. The turbine shroud segment **22** includes a carrier segment **24** arranged circumferentially at least partway around an axis **11** of the gas turbine engine **10**, a blade track segment **26** arranged circumferentially at least partway around the axis **11**, a mount system **28** configured to couple the carrier segment **24** to the blade track segment **26**, and a seal system **30** as shown in FIGS. **2-6**. The seal system **30** is configured to seal gaps between the carrier segment **24** and the blade track segment **26** to prevent or block gases from a gas path **25** of the gas turbine engine **10** from flowing between the carrier segment **24** and the blade track segment **26**.

(15) The blade track segment **26** is a ceramic matrix composite component configured to directly face the high temperatures of the gas path **25** of the gas turbine engine **10** to define a portion of the gas path **25**. The carrier segment **24** is a metallic support component configured to interface with other metallic components of the gas turbine engine **10**, such as the case **17**, to support the blade track segment **26** to radially locate the blade track segment **26** relative to the axis **11**. The mount system **28** includes at least one retainer **80**, **82**, and illustratively the mount system **28** includes two retainers **80**, **82** that each extend axially into the blade track segment **26** and the carrier segment **24** to couple the blade track segment **26** to the carrier segment **24**. The seal system **30** is arranged radially between the carrier segment **24** and the blade track segment **26** to seal a cavity **48** (sometimes referred to as an attachment-receiving space) defined by the carrier segment **24** to block gases from flowing between the carrier segment **24** and the blade track segment **26** and into the cavity **48**.

(16) The seal system **30** includes a forward seal assembly **30F** and an aft seal assembly **30A** as shown in FIGS. **2** and **4-6**. The forward seal assembly **30F** is located radially between the carrier segment **24** and the blade track segment **26** on a forward side of the blade track segment **26**. The aft seal assembly **30A** located radially between the carrier segment **24** and the blade track segment **26** on an aft side of the blade track segment **26**. The forward seal assembly **30F** includes a seal **31** arranged in a first channel **52** formed in the carrier segment **24**. The aft seal assembly **30A** includes a plurality of seal members **38A**, **38B**, **38C** arranged in a second channel **58** formed in the carrier segment **24**.

(17) The forward seal assembly **30F** includes the seal **31** that extends circumferentially about the axis **11** and axially across the first channel **52** formed in the carrier segment **24** as shown in FIGS. **4-6**. The seal **31** comprises a first seal member **32** and a second seal member **34** that each extend circumferentially about the axis **11** as shown in FIG. **4**. Both the first seal member **32** and the second seal member **34** are arranged in the first channel **52** formed in the carrier segment **24**. The first seal member **32** extends radially between the carrier segment **24** and the blade track segment **26**. The second seal member **34** is arranged in a side groove **37G** formed in the first seal member

32 axially between the first seal member **34** and the carrier segment **24** so that the second seal member **34** is engaged with the carrier segment **24**. By arranging the second seal member **34**, the second seal member **34** is coupled to the first seal member **32** for movement therewith.

(18) During operation of a gas turbine engine **10**, the hot, high-pressure products directed into the turbine **18** from the combustor **16** flow across a radially-inwardly opening surface of a shroud wall **70** of the blade track segment **26** that defines a portion of the gas path **25**. The seal system **30** blocks the hot, high-pressure products from flowing into the cavity **48** of the turbine shroud segment **22**. Some turbine shroud assemblies use seals having at least two seal members, where one of the seals is configured to be compressed between the carrier segment **24** and the blade track segment **26** to bias the other seal member(s) into engagement with the shroud wall **70** of the blade track segment **26**, thereby improving the seal therebetween.

(19) In some embodiments, the carrier segment **24** may also include buffer air passageways to direct relatively high-pressure air (sometimes referred to as buffer air) into the channel(s) formed in the carrier segment **24** to distribute the high-pressure air along the seal members. The high-pressure air supplied to the channel(s) is used help keep the gases in the gas path **25** out of the cavity **48** in the event of a seal failure. The high-pressure or buffer air is usually jetted through the seal members arranged in the channel(s), which may cause the seal members to wear, specifically oxidize, significantly reducing the overall life of the seal members and the effectiveness of the seal members.

(20) Instead of using another seal member to urge the other seal member(s) into engagement with the shroud wall **70** of the blade track segment **26**, the forward seal assembly **30F**—also referred to as the buffer air seal assembly **30F**—is pressure-activated. The buffer air is discharged into the first channel **52** formed in the carrier segment **24** as suggested by arrow A in FIG. 6. The buffer air discharged pushes or urges the first seal member **32** of the seal **31** radially inward into engagement with the shroud wall **70** of the blade track segment **26**. The pressure activation of the buffer air seal assembly **30F** causes the first seal member **32** of the seal **31** to be urged radially inward into engagement with the shroud wall **70** of the blade track segment **26**. The seal **31** essentially acts as a piston and is urged radially inward into engagement with the shroud wall **70** of the blade track segment **26** when the buffer air is discharged into the first channel **52**.

(21) The second seal member **34** is coupled with the first seal member **32** for movement therewith so that the second seal member **34** is urged radially inward with the first seal member **32**. The second seal member **34** is located axially between the carrier segment **24** and the first seal member **32** and engages the carrier segment **24**. In the illustrative embodiment, the first seal member **32** and the second seal member **34** of the seal **31** are located in the first channel **52** so that the first seal member **32** engages the carrier segment at the forward side and the second seal member **34** engages the carrier segment at the aft side as shown in FIGS. 5 and 6.

(22) In the illustrative embodiment, the buffer air seal assembly **30F** includes the seal **31** and at least one bias member **36** compressed radially between the carrier segment **24** and the seal **31** to apply a bias force to the seal **31** to bias the seal **31** radially inward towards the shroud wall **70** of the blade track segment **26** as shown in FIGS. 4-6. In some embodiments, the buffer air seal assembly **30F** may include a plurality of bias members **36** spaced apart circumferentially relative to the axis **11** as shown in FIGS. 4-6.

(23) Each of the bias members **36** is arranged radially between the carrier segment **24** and the first seal member **32** of the seal **31** to bias the seal **31** radially inward into engagement with the shroud wall **70** of the blade track segment **26**. The bias member **36** or bias members **36** may be included to assist the pressure activation of the seal **31**. The number of bias members **36** may vary based on the pressure activation of the seal **31**. In other words, the number of bias members **36** may be increased if additional assistance is needed to help urge the seal **31** into engagement with the blade track segment **26**.

(24) In the illustrative embodiment, each of the bias members **36** is a spring. In the illustrative

embodiment, each of the bias members **36** is a coil spring. In some embodiments, each of the bias members **36** may be a leaf spring or a wave spring. In some embodiments, each of the bias members **36** may be another suitable spring type. In some embodiments, each of the bias members **36** may be another suitable type of bias member configured to apply a bias force to the seal **31**.

(25) The buffer air seal assembly **30F** includes the seal **31** and the bias members **36** as shown in FIGS. **4-6**. The seal **31** includes a first seal member **32** that extends into a radially-inwardly opening first channel **52** formed in the carrier segment **24** and a second seal member **34** arranged in the groove **37G** formed in the first seal member **32** so that the second seal member **34** is axially between the first seal member **32** and the carrier segment **24** as shown in FIGS. **5** and **6**. The channel **52** extends circumferentially relative to the axis **11**. The bias member **36** is arranged radially between the seal **31** and the carrier segment **24** to urge the seal **31** radially inward into engagement with the blade track segment **26**.

(26) The first channel **52** is defined by a first end surface **51**, a first—support-wall surface **53**, and a second support-wall surface **55** as shown in FIG. **6**. The first support-wall surface **53** extends radially inward from the first end surface **51**. The second support-wall surface **55** extends radially inward from the first end surface **51** and spaced apart axially from the first—support-wall surface **53**. The first seal member **32** may engage the first support-wall surface **53** and the second seal member **34** is engaged with the second support-wall surface **55** as shown in FIG. **6**.

(27) In the illustrative embodiment, the first seal member **32** also extends into a radially-outwardly opening third channel **78** formed in the blade track segment **26**. The third channel **78** extends circumferentially relative to the axis **11**. The third channel **78** is axially aligned with the first channel **52** and extends the same axial length as the first channel **52** as shown in FIG. **6**. In some embodiments, the first seal member **32** just engages outer surface **70S** of the shroud wall **70** of the blade track segment **26** without any additional channels or grooves formed in the shroud wall **70** of the blade track segment **26**.

(28) The third channel **78** is defined by a third end surface **71**, a first shroud-wall surface **73**, and a second shroud-wall surface **75** as shown in FIG. **6**. The first shroud-wall surface **73** extends radially outward from the third end surface **71**. The second shroud-wall surface **75** extends radially outward from the third end surface **71**. The second shroud-wall surface **75** is spaced apart axially from the first shroud-wall surface **73**. The first seal member **32** is urged into engagement with the third end surface **71** and may engage the first shroud-wall surface **73** as shown in FIG. **6**.

(29) The buffer air passageway **50A** extends radially into a forward support wall **44** included in the carrier segment **24** and opens into the first channel **52**. The buffer air passageway **50A** discharges the buffer air into the buffer air cavity **49** defined between the surfaces **51**, **53**, **55** of the first channel **52** and the first and second seal members **32**, **34** as shown in FIGS. **5** and **6**. The buffer air discharged pressurizes the buffer air cavity **49** to urge the first seal member **32** radially inward into engagement with in the blade track segment **26**.

(30) The second seal member **34** comprises a single strand of solid metallic material. In the illustrative embodiment, the second seal member **34** is a wire seal. In some embodiments, the second seal member **34** may be another suitable seal type. In the illustrative embodiment, the second seal member **34** prevents the buffer air from leaking around the first seal member **32** between the first seal member **32** and the carrier segment **24**.

(31) The first seal member **32** has a rectangular cross-sectional shape when viewed in the circumferential direction relative to the axis **11** as shown in FIGS. **5** and **6**. In some embodiments, the first seal member **32** may have another suitable seal type.

(32) The rectangular cross-sectional shape of the first seal member **32** defines an inner surface **33**, and outer surface **35**, and two side surfaces **37**, **39** as shown in FIGS. **5** and **6**. The inner surface **33** is engaged with the shroud wall **70** of the blade track segment **26**. The outer surface **35** is spaced apart radially from the inner surface **33**. The side surfaces **37**, **39** are spaced apart axially from one another. Each side surface **37**, **39** extends radially between the inner surface **33** and the outer

surface 35.

(33) The first seal member 32 is formed to include a side groove 37G, an inner groove 33G, and at least one through hole 32H as shown in FIGS. 4-6. The side groove 37G extends axially into the first side surface 37 of the first seal member 32 and circumferentially relative to the axis 11. The inner groove 33G extends radially into the inner surface 33 of the first seal member 32 and circumferentially relative to the axis 11. The second seal member 34 is arranged in the side groove 37G and engages the carrier segment 24 in the first channel 52. The through hole 32H extends radially through the first seal member 32 and opens into the inner groove 33G to allow the buffer air to flow through the first seal member 32 into the inner groove 33G.

(34) In the illustrative embodiment, the first seal member 32 is formed to include a plurality of through holes 32H as shown in FIG. 4. The through holes 32H are spaced apart circumferentially relative to the axis 11. In some embodiments, the number of through holes 32H may be varied to control the amount of buffer air flowing to the inner groove 33G.

(35) Each of the channels 52, 78 also have a rectangular cross-sectional shape when viewed in the circumferential direction relative to the axis 11. The rectangular shape of the channels 52, 78 defines the respective surfaces 51, 51, 53, 55, 71, 73, 75. In the illustrative embodiment, the depth of the first channel is greater than the depth of the third channel as shown in FIGS. 5 and 6. In some embodiments, the depths of the channels may all be equal.

(36) With the seal system 30 of the present disclosure initially described above, the gas turbine engine 10 is now described in more detail. The gas turbine engine 10 includes a fan 12, a compressor 14, a combustor 16, and a turbine 18 as shown in FIG. 1. The fan 12 is driven by the turbine 18 and provides thrust for propelling an air vehicle. The compressor 14 compresses and delivers air to the combustor 16. The combustor 16 mixes fuel with the compressed air received from the compressor 14 and ignites the fuel. The hot, high-pressure products of the combustion reaction in the combustor 16 are directed into the turbine 18 to cause the turbine 18 to rotate about an axis 11 and drive the compressor 14 and the fan 12. In some embodiments, the fan may be replaced with a propeller, drive shaft, or other suitable configuration.

(37) The turbine 18 includes at least one turbine wheel assembly 19 and a turbine shroud 20 positioned to surround the turbine wheel assembly 19 as shown in FIGS. 1 and 2. The turbine wheel assembly 19 includes a plurality of blades 21 coupled to a rotor disk 23 for rotation with the disk 23. The hot, high pressure combustion products from the combustor 16 are directed toward the blades 21 of the turbine wheel assemblies 19 along the gas path 25. The turbine shroud 20 is coupled to the outer case 17 of the gas turbine engine 10 and extends around the turbine wheel assembly 19 to block gases from passing over the turbine blades 21 during use of the turbine 18 in the gas turbine engine 10.

(38) In the illustrative embodiment, the turbine shroud 20 is made up of a number of turbine shroud segment assemblies 22 that each extend circumferentially partway around the axis 11 and cooperate to surround the turbine wheel assembly 19. In other embodiments, the turbine shroud 20 is annular and non-segmented to extend fully around the axis 11 and surround the turbine wheel assembly 19. In yet other embodiments, certain components of the turbine shroud 20 are segmented while other components are annular and non-segmented.

(39) Each turbine shroud segment 22 includes the carrier segment 24, blade track segment, the mount system 28, and the seal system 30 as shown in FIGS. 2-6. The carrier segment 24 and the blade track segment 26 are arranged circumferentially partway about the axis 11. The blade track segment 26 includes the shroud wall 70 that extends circumferentially partway around the axis 11 to define a portion of the gas path 25 and an attachment feature 72 that extends radially from the shroud wall 70 into the cavity 48 of the carrier segment 24. The mount system 28 is configured to couple the blade track segment 26 to the carrier segment 24. The seal system 30 is arranged radially between the carrier segment 24 and the blade track segment 26 to seal gaps therebetween.

(40) The carrier segment 24 includes an outer wall 40, a pair of hangers 42, a forward support wall

44, and an aft support wall **46** as shown in FIGS. 3-6. The outer wall **40** extends circumferentially at least partway about the axis **11**. The hangers **42** extend radially outward from the outer wall **40** and engage the case **17** to couple the turbine shroud segment **22** to the rest of the engine **10**. The forward support wall **44** extends radially inward from the outer wall **40** at a forward end of the outer wall **40** axially forward of the attachment feature **72** and the aft support wall **46** extends radially inward from the outer wall **40** at an aft end of the outer wall **40** axially aft of the attachment feature **72**.

(41) In the illustrative embodiment, the carrier segment **24** further includes a first intermediate support wall **45** and a second intermediate support wall **47** as shown in FIGS. 3, 4, and 6. The first intermediate support wall **45** and the second intermediate support wall **47** each extend radially inward from the outer wall **40** of the carrier segment **24** axially between the first and second support walls **44**, **46**. The second intermediate support wall **47** is spaced apart axially from the first intermediate support wall **45** in the illustrative embodiment.

(42) The forward and aft support walls **44**, **46** of the carrier segment **24** each include corresponding channels **52**, **58** as shown in FIGS. 5 and 6. The forward support wall **44** is formed to include the first channel **52**. The aft support wall **46** is formed to include a second channel **58** in the illustrative embodiment. Both the first channel **52** and the second channel **58** are radially-inwardly opening so that the channels **52**, **58** are open towards the blade track segment **26**.

(43) In the illustrative embodiment, only the forward support wall **44** includes the buffer air passageway **50A** as shown in FIGS. 5 and 6. The forward support wall **44** includes at least one discrete buffer air passageway **50A** that extends radially into the forward support wall **44** and opens into the first channel **52**. In the illustrative embodiment, the forward support wall **44** includes a plurality of buffer air passageways **50A** that are spaced apart circumferentially about the axis **11**.

(44) In the illustrative embodiment, each buffer air passageway **50A** extends from an outer cavity **60A**, **60B**, **60C** formed in the outer wall **40** of the carrier segment **24** as shown in FIGS. 5 and 7. The buffer air flows from the outer cavity **60A**, **60B**, **60C** to the buffer air passageways **50A**. In some embodiments, the carrier segment **24** is formed to include an outer cavity **60A**, **60B**, **60C** for each buffer air passageway **50A** that supplies the corresponding buffer air passageway **50A** with buffer air.

(45) The first channel **52** formed in the forward support wall **44** is defined by the first end surface **51**, the first support-wall surface **53**, and a second support-wall surface **55** as shown in FIG. 6. The first support-wall surface **53** extends radially inward from the first end surface **51**. The second support-wall surface **55** extends radially inward from the first end surface **51**. The second support-wall surface **55** is spaced apart axially from the first support-wall surface **53**.

(46) The blade track segment **26** includes the shroud wall **70** and the attachment feature **72** as shown in FIGS. 4-6. The shroud wall **70** that extends circumferentially partway around the axis **11**. The attachment feature **72** includes a first attachment flange **74** and a second attachment flange **76** that each extend radially outward from the shroud wall **70**. The second attachment flange **76** is spaced apart axially from the first attachment flange **74**.

(47) In the illustrative embodiment, the forward support wall **44** extends radially inward from the outer wall **40** axially forward of the first attachment flange **74** of the blade track segment **26**. The aft support wall **46** extends radially inward from the outer wall **40** axially aft of the second attachment flange **76** of the blade track segment **26**. The first intermediate support wall **45** extends radially inward from the outer wall **40** axially aft of the first attachment flange **74** so that the first attachment flange **74** is axially between the forward support wall **44** and the first intermediate support wall **45**. The second intermediate support wall **47** extends radially inward from the outer wall **40** axially forward of the second attachment flange **76** of the blade track segment **26** so that the second attachment flange **76** is located axially between the aft support wall **46** and the second intermediate support wall **47**.

(48) In the illustrative embodiment, the shroud wall **70** of the blade track segment **26** is formed to

include the third channel **78** as shown in FIGS. **4-6**. The third channel **78** is defined by a third end surface **71**, a first shroud-wall surface **73**, and a second shroud-wall surface **75** as shown in FIG. **6**. The first shroud-wall surface **73** extends radially outward from the third end surface **71**. The second shroud-wall surface **75** extends radially outward from the third end surface **71**. The second shroud-wall surface **75** is spaced apart axially from the first shroud-wall surface **73**.

(49) In some embodiments, the shroud wall **70** of the blade track segment **26** may have a coating layer applied in the third channel **78**. The coating layer in the third channel **78** may define the third end surface **71**, the first shroud-wall surface **73**, and the second shroud-wall surface **75**. The first seal member **32** may engage the coating layer in the third channel **78**.

(50) The mount system **28** includes at least one retainer **80, 82**, illustratively two retainers **80, 82** that each extend axially into the blade track segment **26** and the carrier segment **24** to couple the blade track segment **26** to the carrier segment **24**. The retainers **80, 82** extend axially into the forward support wall **44**, through the first attachment flange **74**, the intermediate support walls **43, 45**, and the second attachment flange **76**, and into the aft support wall **46** of the carrier segment **24** so as to couple the blade track segment **26** to the carrier segment **24**.

(51) In the illustrative embodiment, the mount system **28** includes the retainers **80, 82** and corresponding retainer plugs **84, 86** as shown in FIG. **4**. Each of the retainer plugs **84** extends into an installation apertures formed in the aft support wall **46** to block removal of the corresponding retainers **80, 82** through the installation apertures in the carrier segment **24**.

(52) In the illustrative embodiment, the retainers **80, 82** are both split pins as shown in FIG. **3**. Each retainer **80, 82** includes a first pin **80A, 82A** and a second pin **80B, 82B** arranged axially aft of the first pin **80A, 82A** as shown in FIG. **4**.

(53) A method of assembling and using the turbine shroud segment **22** may include several steps. The method includes arranging the seal members **32, 34** of the forward seal assembly **30F** and the seal members **38A, 38B, 38C** of the aft seal assembly **30A** in the corresponding channels **52, 58** before arranging the blade track segment **26** adjacent to the carrier segment **24**. The method includes arranging the seal **31** in the first channel **52** and arranging the other seal members **38A, 38B, 38C** in the second channel **58**. The seal assemblies **30F, 30A** may be arranged in the corresponding channels **52, 58**, in any order.

(54) The step of arranging the seal **31** in the first channel **52** may include arranging the second seal member **34** in the side groove **37G** formed in the first seal member **32** and then arranging the assembled first and second seal members **32, 34** in the first channel **52** formed in the carrier segment **24** after arranging the second seal member **34** in the side groove **37G** of the first seal member **32**. In illustrative embodiment, the method includes arranging at least one bias member **36** in the first channel **52** before arranging the seal **31** in the first channel **52**.

(55) Once all the seal members **32, 34, 38A, 38B, 38C** are arranged in the corresponding channels **52, 58**, the blade track segment **26** is arranged adjacent to the carrier segment **24** so that the seal **31** of the forward seal assembly **30F** and the seal members **38A, 38B, 38C** of the aft seal assembly **30A** are radially between the carrier segment **24** and the shroud wall **70** of the blade track segment **26** to block gases in the gas path from flowing between the carrier segment **24** and the blade track segment **26**. The blade track segment **26** is arranged adjacent to the carrier segment **24** so that the first attachment flange **74** and the second attachment flange **76** of the attachment feature **72** extend into sections of the cavity **48**. In the illustrative embodiment, the blade track segment **26** is arranged adjacent to the carrier segment **24** so that the first seal member **32** of the seal **31** also extends into the third channel **78** formed in the shroud wall **70** of the blade track segment **26**.

(56) In some embodiments, the method further includes inserting the retainers **80, 82** into the carrier segment **24** and the blade track segment **26** to couple the blade track segment **26** to the carrier segment **24**. The method includes inserting one retainer **80** axially into the carrier segment **24** and through the attachment feature **72** of the blade track segment **26** to couple the blade track segment **26** to the carrier segment **24**. The method further includes inserting another retainer **82**

axially into the carrier segment **24** and through the attachment feature **72** of the blade track segment **26** to couple the blade track segment **26** to the carrier segment **24**. The second retainer **82** is inserted at a location spaced apart circumferentially from the first retainer **80**.

(57) The method further includes discharging a flow of buffer air through the at least one buffer air passageway **50A**. The flow of buffer air is suggested by arrow **A**. The method may further include discharging the flow of buffer air through the plurality of buffer air passageways **50A**. The method includes discharging the flow of buffer air through the buffer air passageway **50A** radially inward away from the carrier segment **24** into the buffer air cavity **49** defined between the seal **31** and the first channel **52** formed in the carrier segment **24**. The buffer air is discharged into the buffer air cavity **49** to urge the seal **31** radially inward into engagement with the blade track segment **26**.

(58) The buffer air discharged into the buffer air cavity **49** and into the inner groove **33G** establishes a higher pressure $P_{sub.1}$ in the buffer air cavity **49** than the pressure $P_{sub.2}$ in the region axially forward of the buffer air seal assembly **30F** and the pressure $P_{sub.3}$ in the cavity **48** as shown in FIG. **6**. The pressure $P_{sub.3}$ in the cavity **48** is lower than the pressure $P_{sub.2}$ in the region axially forward of the buffer air seal assembly **30F** radially outward of the gas path **25** as shown in FIG. **6**. The buffer air may be provided from the compressor **14** of the gas turbine engine **10**.

(59) The through holes **32H** in the first seal member **32** allow a portion of the buffer air to flow through the first seal member **32** into the inner groove **33G**. The buffer air flowing into the buffer air cavity **49** may leak around the sides of the seal **31** in the illustrative embodiment. The buffer air flowing into the inner groove **33G** may leak around the end of the first seal member **32** as the pressure $P_{sub.3}$ in the cavity **48** and the pressure $P_{sub.2}$ in the region axially forward of the buffer air seal assembly **30F** are at a lower pressure.

(60) Another embodiment of a turbine shroud segment **222** in accordance with the present disclosure is shown in FIGS. **7-9**. The turbine shroud segment **222** is substantially similar to the turbine shroud segment **22** shown in FIGS. **1-7** and described herein. Accordingly, similar reference numbers in the **200** series indicate features that are common between the turbine shroud segment **22** and the turbine shroud segment **222**. The description of the turbine shroud segment **22** is incorporated by reference to apply to the turbine shroud segment **222**, except in instances when it conflicts with the specific description and the drawings of the turbine shroud segment **22**.

(61) The turbine shroud segment **222** includes a carrier segment **224** arranged circumferentially at least partway around an axis **11** of the gas turbine engine **10**, a blade track segment **226** arranged circumferentially at least partway around the axis **11**, and a seal system **230** as shown in FIGS. **7-9**. The seal system **230** is configured to seal gaps between the carrier segment **224** and the blade track segment **226** to prevent or block gases from the gas path **25** of the gas turbine engine **10** from flowing between the carrier segment **224** and the blade track segment **226**.

(62) The seal system **230** includes forward seal assembly **230F** located radially between the carrier segment **224** and the blade track segment **226** on a forward side of the blade track segment **226** and an aft seal assembly **230A** located radially between the carrier segment **224** and the blade track segment **226** on an aft side of the blade track segment **226**. The forward seal assembly **230F**—also referred to as the buffer air seal assembly **230F**—includes a seal **231** having a plurality of seal members **232A**, **232B**, **232C** and bias members **236**.

(63) The seal **231** includes an outer seal member **232A**, a forward seal member **232B**, and an aft seal member **232C** as shown in FIGS. **7-9**. Each of the seal members **232A**, **232B**, **232C** extend circumferentially relative to the axis **11**. The outer seal member **232A**, the forward seal member **232B**, and aft seal member **232C** are free to move relative to each other. When arranged in the first channel **52**, the forward seal member **232B** and the aft seal member **232C** each extend radially inward from the outer seal member **232A**. The aft seal member **232C** is spaced apart axially from the forward seal member **232B** to define an inner groove **233G** therebetween.

(64) The outer seal member **232A** is formed to include at least one through hole **232H** as shown in

FIGS. 7-9. The through hole 232H extends radially through the outer seal member 232A and opens into the inner groove 233G to allow the buffer air to flow through the outer seal member 232A into the inner groove 233G to urge the forward seal member 232B axially forward and the aft seal member 232C axially aft into engagement with the carrier segment 224 and the blade track segment 226. In the illustrative embodiment, the outer seal member 232A is formed to include a plurality of through holes 232H as shown in FIGS. 7 and 9. The through holes 232H are spaced apart circumferentially relative to the axis 11.

(65) In the illustrative embodiment, the buffer air seal assembly 230F includes the seal 231 and at least one bias member 236 compressed radially between the carrier segment 224 and the seal 231 to apply a bias force to the seal 231 to bias the seal 231 radially inward towards the shroud wall 270 of the blade track segment 226 as shown in FIGS. 7-9. In some embodiments, the buffer air seal assembly 230F may include a plurality of bias members 236 spaced apart circumferentially relative to the axis 11 as shown in FIGS. 7-9.

(66) Each of the bias members 236 is arranged radially between the carrier segment 224 and the outer seal member 232A of the seal 231 to bias the forward and aft seal members 232B, 232C of the seal 231 radially inward into engagement with the shroud wall 270 of the blade track segment 226. The bias member 236 or bias members 236 may be included to assist the pressure activation of the seal 231. The forward and aft seal members 232B, 232C are radially inward of the outer seal member 232A so that the bias members 236 urge the outer seal member 232A radially inward to cause the forward and aft seal members 232B, 232C to also be urged radially inward into engagement with the blade track segment 226.

(67) The first channel 252 is defined by a first end surface 251, a first—support-wall surface 253, and a second support-wall surface 255 as shown in FIG. 9. The first support-wall surface 253 extends radially inward from the first end surface 251. The second support-wall surface 255 extends radially inward from the first end surface 251 and spaced apart axially from the first support-wall surface 253. The forward seal member 232B engages the first support-wall surface 53 and the aft seal member 232C engages the second support-wall surface 55 as shown in FIG. 9.

(68) In the illustrative embodiment, the forward and aft seal members 232B, 232C also extend into a radially-outwardly opening third channel 278 formed in the blade track segment 226. The third channel 278 extends circumferentially relative to the axis 11. The third channel 278 is axially aligned with the first channel 252 and extends the same axial length as the first channel 252.

(69) The third channel 278 is defined by a third end surface 271, a first shroud-wall surface 273, and a second shroud-wall surface 275 as shown in FIG. 9. The first shroud-wall surface 273 extends radially outward from the third end surface 271. The second shroud-wall surface 275 extends radially outward from the third end surface 271. The second shroud-wall surface 275 is spaced apart axially from the first shroud-wall surface 273.

(70) The forward and aft seal members 232B, 232C are both urged into engagement with the third end surface 71 as shown in FIG. 9. The forward seal member 232B engages the first shroud-wall surface 273 and the aft seal member 232C engages the second shroud-wall surface 275.

(71) The buffer air passageway 250A extends radially into the forward support wall 244 included in the carrier segment 224 and opens into the first channel 252. The buffer air passageway 250A discharges the buffer air into the buffer air cavity 249 defined between the surfaces 251, 253, 255 of the first channel 252 and the outer seal member 232A as shown in FIGS. 8 and 9. The buffer air discharged pressurizes the buffer air cavity 249 to urge the seal 231 radially inward into engagement with the blade track segment 226.

(72) Each of the seal members 232A, 232B, 232C have a rectangular cross-sectional shape when viewed in the circumferential direction relative to the axis 11 as shown in FIGS. 7-9. In some embodiments, the seal members 232A, 232B, 232C may have another suitable seal type.

(73) The rectangular cross-sectional shape of the seal members 232A, 232B, 232C defines the surfaces that engage the carrier segment 224 and/or the blade track segment 226. Each side surface

237, 239 of the respective seal member **232B, 232C** engages the respective surfaces of the carrier segment **224** and the blade track segment **226**.

(74) The carrier segment **224** includes an outer wall **240**, a pair of hangers **242**, a forward support wall **244**, and an aft support wall **246** as shown in FIGS. 7-9. The outer wall **240** extends circumferentially at least partway about the axis **11**. The forward support wall **244** extends radially inward from the outer wall **240** at a forward end of the outer wall **240** axially forward of the attachment flange **274** and the aft support wall **246** extends radially inward from the outer wall **240** at an aft end of the outer wall **240** axially aft of the attachment flange **276**.

(75) The forward support wall **244** of the carrier segment **224** includes the first channel **252** as shown in FIGS. 8 and 9. The forward support wall **244** is formed to include the first channel **252**. The first channel **252** is radially-inwardly opening so that the channel **252** is open toward the blade track segment **226**. In the illustrative embodiment, only the forward support wall **244** includes the buffer air passageway **250A** as shown in FIGS. 8 and 9.

(76) The blade track segment **226** includes the shroud wall **270** and the attachment flanges **274, 276** as shown in FIGS. 7-9. The shroud wall **270** that extends circumferentially partway around the axis **11**. The first attachment flange **274** and the second attachment flange **276** each extend radially outward from the shroud wall **270**. The second attachment flange **276** is spaced apart axially from the first attachment flange **274**.

(77) In the illustrative embodiment, the shroud wall **270** of the blade track segment **226** is formed to include the third channel **278** as shown in FIGS. 7-9. The third channel **278** is defined by a third end surface **271**, a first shroud-wall surface **273**, and a second shroud-wall surface **275** as shown in FIG. 9. The first shroud-wall surface **273** extends radially outward from the third end surface **271**. The second shroud-wall surface **275** extends radially outward from the third end surface **271**. The second shroud-wall surface **275** is spaced apart axially from the first shroud-wall surface **273**.

(78) A method of assembling and using the turbine shroud segment **222** may include several steps. The method includes arranging the seal members **232A, 232B, 232C** of the seal **231** of the forward seal assembly **230F** in the first channel **252** before arranging the blade track segment **226** adjacent to the carrier segment **224**. In the illustrative embodiment, the method includes arranging the bias member **236** or bias members **236** in the first channel **252** before arranging the seal **231** in the first channel **252**.

(79) The step of arranging the seal **231** in the first channel **252** may include arranging the outer seal member **232A** in the first channel **252** so that the bias member **236** is located radially between the carrier segment **224** and the outer seal member **232A**. Then the forward and aft seal members **232B, 232C** may be arranged in the first channel **252** after arranging the outer seal member **232A** in the first channel, **252**. The ends of the seal members **232B, 232C** may be engages with the outer seal member **232A** when arranging the forward and aft seal members **232B, 232C** in the first channel **252**. Additionally, the method arranging step further includes spacing the aft seal member **232C** apart axially from the forward seal member **232B** to define the inner groove **233G** therebetween.

(80) Once all the seal members **232A, 232B, 232C** are arranged in the first channel **252**, the blade track segment **226** is arranged adjacent to the carrier segment **224** so that the seal **231** of the forward seal assembly **230F** is radially between the carrier segment **224** and the shroud wall **270** of the blade track segment **226** to block gases in the gas path from flowing between the carrier segment **224** and the blade track segment **226**. In the illustrative embodiment, the blade track segment **226** is arranged adjacent to the carrier segment **224** so that the forward and aft seal members **232B, 232C** of the seal **231** also extend into the third channel **278** formed in the shroud wall **270** of the blade track segment **226**.

(81) The method further includes discharging a flow of buffer air through the at least one buffer air passageway **250A**. The flow of buffer air is suggested by arrow A. The method may further include discharging the flow of buffer air through the plurality of buffer air passageways **250A**. The

method includes discharging the flow of buffer air through the buffer air passageway **250A** radially inward away from the carrier segment **224** into the buffer air cavity **249** defined between the seal **231** and the first channel **252** formed in the carrier segment **224**. The buffer air is discharged into the buffer air cavity **249** to urge the seal **231** radially inward into engagement with the blade track segment **226**.

(82) The through holes **232H** in the first seal member **232** allow a portion of the buffer air to flow through the first seal member **232** into the inner groove **233G**. The buffer air may pressurize the buffer air cavity **249** to establish a pressure $P_{sub.4}$ in the inner groove **233G** that is greater than the pressure $P_{sub.2}$ in the region axially forward of the buffer air seal assembly **230F** and the pressure $P_{sub.3}$ in the region axially aft of the buffer air seal assembly **230F**.

(83) The buffer air discharged into the buffer air cavity **249** and into the inner groove **233G** establishes a higher pressure $P_{sub.1}$ in the buffer air cavity **249** than the pressure $P_{sub.2}$ in the region axially forward of the buffer air seal assembly **230F** and the pressure $P_{sub.3}$ in the cavity **248** as shown in FIG. 9. The pressure $P_{sub.3}$ in the cavity **248** is lower than the pressure $P_{sub.2}$ in the region axially forward of the buffer air seal assembly **230F** as shown in FIG. 9. The buffer air may be provided from the compressor **14** of the gas turbine engine **10**.

(84) The through holes **232H** in the outer seal member **232A** allow a portion of the buffer air to flow through the first seal member **232** into the inner groove **233G**. The pressure $P_{sub.4}$ in the inner groove **233G** is greater than the pressure $P_{sub.3}$ in the cavity **48** and the pressure $P_{sub.2}$ in the region axially forward of the buffer air seal assembly **30F**, which may urge the forward seal member **232B** axially forward into engagement with the surfaces **253**, **273** and may urge the aft seal member **232C** axially aft into engagement with the surfaces **255**, **275**. The buffer air flowing into the inner groove **233G** may leak around the ends of the seal members **232B**, **232C** as the pressure $P_{sub.3}$ in the cavity **48** and the pressure $P_{sub.2}$ in the region axially forward of the buffer air seal assembly **30F** are at a lower pressure.

(85) Another embodiment of a turbine shroud segment **322** in accordance with the present disclosure is shown in FIGS. 10-12. The turbine shroud segment **322** is substantially similar to the turbine shroud segment **22** shown in FIGS. 1-7 and described herein. Accordingly, similar reference numbers in the **300** series indicate features that are common between the turbine shroud segment **22** and the turbine shroud segment **322**. The description of the turbine shroud segment **22** is incorporated by reference to apply to the turbine shroud segment **322**, except in instances when it conflicts with the specific description and the drawings of the turbine shroud segment **22**.

(86) The turbine shroud segment **322** includes a carrier segment **324** arranged circumferentially at least partway around an axis **11** of the gas turbine engine **10**, a blade track segment **326** arranged circumferentially at least partway around the axis **11**, and a seal system **330** as shown in FIGS. 10-12. The seal system **330** is configured to seal gaps between the carrier segment **324** and the blade track segment **326** to prevent or block gases from the gas path **25** of the gas turbine engine **10** from flowing between the carrier segment **324** and the blade track segment **326**.

(87) The seal system **330** includes forward seal assembly **330F** located radially between the carrier segment **324** and the blade track segment **326** on a forward side of the blade track segment **326** and an aft seal assembly **330A** located radially between the carrier segment **324** and the blade track segment **326** on an aft side of the blade track segment **326**. The forward seal assembly **330F**—also referred to as the buffer air seal assembly **330F**—includes a seal **331** having a plurality of seal members **332**, **334** and bias members **336**.

(88) The seal **331** includes a first seal member **332** and a second seal member **334** as shown in FIGS. 10-12. Each of the first and second seal members **332**, **334** extend circumferentially relative to the axis **11**. The second seal member **334** is arranged around the first seal member **332** so that the second seal member **334** is coupled with the first seal member **332**. The seal **331** is pressure activated such that the first seal member **332** is urged radially inward, which causes the second seal member **334** to be urged into engage with the blade track segment **226**.

(89) The first seal member **332** has a rectangular cross-sectional shape when viewed in the circumferential direction as shown in FIGS. **10-12**. The rectangular cross-sectional shape of the first seal member **332** defines an inner surface **333**, an outer surface **335** spaced apart radially from the inner surface **333**, and two side surfaces **337**, **339** spaced apart axially that extend radially between the inner surface **333** and the outer surface **335**. The second seal member **334** extends around a portion of the first seal member **332** so that the second seal member **334** engages the carrier segment **324** and the blade track segment **326** when the seal **331** is in the first channel **352**.

(90) The second seal member **334** includes an inner band **388**, a first attachment arm **390**, and a second attachment arm **392** as shown in FIGS. **10** and **12**. The inner band **388** is arranged radially between the first seal member **332** and the shroud wall **370** of the blade track segment **326**. The first attachment arm **390** extends from the inner band **388** around one side **339** of the first seal member **332** and the second attachment arm **392** extends from the inner band **388** opposite the first attachment arm **390** around the other side **337** of the first seal member **332**. The first attachment arm **390** and the second attachment arm **392** may each extend into the first seal member **332** to couple the second seal member **334** with the first seal member **332**.

(91) The first attachment arm **390** extends radially outward and axially forward from the inner band **388** into engagement with the carrier segment **324**. The first attachment arm **390** then bends back toward the first seal member **332** as shown in FIGS. **10-12**. Similarly, the second attachment arm **392** extends radially outward and axially aft from the inner band **388** into engagement with the carrier segment **324**. The second attachment arm **392** then bends back toward the first seal member **332** as shown in FIGS. **10-12**. The inner band **388** is engaged with the shroud wall **370** of the blade track segment **326**.

(92) The first seal member **332** is formed to include an inner groove **333G** and at least one through hole **332H** as shown in FIGS. **10-12**. The inner groove **333G** extends radially into the inner surface **333** of the first seal member **332** and circumferentially relative to the axis **11**. The through hole **332H** extends radially through the first seal member **332** and opens into the inner groove **333G** to allow the buffer air to flow through the first seal member **332** into the inner groove **333G**. In the illustrative embodiment, the first seal member **332** is formed to include a plurality of through holes **332H** as shown in FIG. **10**. The through holes **332H** are spaced apart circumferentially relative to the axis **11**.

(93) In the illustrative embodiment, the buffer air seal assembly **330F** includes the seal **331** and at least one bias member **336** compressed radially between the carrier segment **324** and the seal **331** to apply a bias force to the seal **331** to bias the seal **331** radially inward towards the shroud wall **370** of the blade track segment **326** as shown in FIGS. **10-12**. In some embodiments, the buffer air seal assembly **330F** may include a plurality of bias members **336** spaced apart circumferentially relative to the axis **11** as shown in FIGS. **10-12**.

(94) Each of the bias members **336** is arranged radially between the carrier segment **324** and the first seal member **332** of the seal **331** to bias the seal **331** radially inward into engagement with the shroud wall **370** of the blade track segment **326**. The bias member **336** or bias members **336** may be included to assist the pressure activation of the seal **331**. The bias members **336** is located radially between the first seal member **332** and the carrier segment **324** to bias the first seal member **332** radially inward to cause the second seal member **334** to be urged radially inward into engagement with the blade track segment **326**.

(95) The first channel **352** is defined by a first end surface **351**, a first—support-wall surface **353**, and a second support-wall surface **355** as shown in FIG. **12**. The first support-wall surface **353** extends radially inward from the first end surface **351**. The second support-wall surface **355** extends radially inward from the first end surface **351** and spaced apart axially from the first support-wall surface **353**. The first attachment arm **390** of the second seal member **334** engages the first support-wall surface **53** and the second attachment arm **392** of the second seal member **334** engages the second support-wall surface **55** as shown in FIG. **12**.

(96) The buffer air passageway **350A** extends radially into the forward support wall **344** included in the carrier segment **324** and opens into the first channel **352**. The buffer air passageway **350A** discharges the buffer air into the buffer air cavity **349** defined between the surfaces **351**, **353**, **355** of the first channel **352** and the outer seal member **332A** as shown in FIGS. **11** and **12**. The buffer air discharged pressurizes the buffer air cavity **349** to urge the seal **331** radially inward into engagement with the blade track segment **326**.

(97) The carrier segment **324** includes an outer wall **340**, a pair of hangers **342**, a forward support wall **344**, and an aft support wall **346** as shown in FIGS. **10-12**. The outer wall **340** extends circumferentially at least partway about the axis **11**. The forward support wall **344** extends radially inward from the outer wall **340** at a forward end of the outer wall **340** axially forward of the attachment flange **374** and the aft support wall **346** extends radially inward from the outer wall **340** at an aft end of the outer wall **340** axially aft of the attachment flange **376**.

(98) The forward support wall **344** of the carrier segment **324** includes the first channel **352** as shown in FIGS. **11** and **12**. The forward support wall **344** is formed to include the first channel **352**. The first channel **352** is radially-inwardly opening so that the channel **352** is open toward the blade track segment **326**. In the illustrative embodiment, only the forward support wall **344** includes the buffer air passageway **350A** as shown in FIGS. **11** and **12**.

(99) The blade track segment **326** includes the shroud wall **370** and the attachment flanges **374**, **376** as shown in FIGS. **10-12**. The shroud wall **370** that extends circumferentially partway around the axis **11**. The first attachment flange **374** and the second attachment flange **376** each extend radially outward from the shroud wall **370**. The second attachment flange **376** is spaced apart axially from the first attachment flange **374**.

(100) A method of assembling and using the turbine shroud segment **322** may include several steps. The method includes arranging the seal members **332A**, **332B**, **332C** of the seal **331** of the forward seal assembly **330F** in the first channel **352** before arranging the blade track segment **326** adjacent to the carrier segment **324**. In the illustrative embodiment, the method includes arranging the bias member **336** or bias members **336** in the first channel **352** before arranging the seal **331** in the first channel **352**.

(101) The step of arranging the seal **331** in the first channel **352** may include arranging the second seal member **334** around the first seal member **332** before arranging both in the first channel **352** so that the bias member **336** is located radially between the carrier segment **324** and the first seal member **332**. The seal **332** is arranged in the first channel **352** so that the attachment arms **390**, **392** of the second seal member **334** engage the surfaces **353**, **355** that define part of the first channel **352**.

(102) Once the seal **331** is arranged in the first channel **352**, the blade track segment **326** is arranged adjacent to the carrier segment **324** so that the seal **331** of the forward seal assembly **330F** is radially between the carrier segment **324** and the shroud wall **370** of the blade track segment **326** to block gases in the gas path from flowing between the carrier segment **324** and the blade track segment **326**. In the illustrative embodiment, the blade track segment **326** is arranged adjacent to the carrier segment **324** so that the inner band **388** of the second seal member **334** engages the shroud wall **370** of the blade track segment **326**.

(103) The method further includes discharging a flow of buffer air through the at least one buffer air passageway **350A**. The flow of buffer air is suggested by arrow A. The method may further include discharging the flow of buffer air through the plurality of buffer air passageways **350A**. The method includes discharging the flow of buffer air through the buffer air passageway **350A** radially inward away from the carrier segment **324** into the buffer air cavity **349** defined between the seal **331** and the first channel **352** formed in the carrier segment **324**. The buffer air is discharged into the buffer air cavity **349** to urge the seal **331** radially inward into engagement with the blade track segment **326**. The through holes **332H** in the first seal member **332** allow a portion of the buffer air to flow through the first seal member **332** into the inner groove **333G**.

(104) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

Claims

1. A turbine shroud assembly for use with a gas turbine engine, the turbine shroud assembly comprising: a carrier segment arranged circumferentially at least partway around an axis, the carrier segment including an outer wall, a first support wall that extends radially inward from the outer wall, and a second support wall that extends radially inward from the outer wall at a location spaced apart axially from the first support wall to define an attachment-receiving space, and the first support wall formed to include a radially-inwardly opening first channel that extends circumferentially relative to the axis and at least one buffer air passageway that extends radially into the forward support wall and opens into the first channel, a blade track segment arranged circumferentially at least partway around the axis to define a portion of a gas path of the gas turbine engine, the blade track segment having a shroud wall that extends circumferentially partway around the axis and an attachment feature that extends radially outward from the shroud wall into the attachment-receiving space formed in the carrier segment, and a buffer air seal assembly arranged in the first channel between the forward support wall of the carrier segment and the shroud wall of the blade track segment to block gases from flowing between the carrier segment and the blade track segment, the buffer air seal assembly including a seal that extends circumferentially relative to the axis and axially across the first channel, wherein the at least one buffer air passageway is configured to discharge buffer air radially inward away from the carrier segment into the first channel to urge the seal radially inward into engagement with the shroud wall of the blade track segment, wherein the buffer air seal assembly further includes a bias member compressed radially between the carrier segment and the seal to apply a bias force to the seal to bias the seal radially inward towards the shroud wall of the blade track segment, and wherein the seal is formed to include an inner groove that extends radially inward into the seal and circumferentially relative to the axis and at least one through hole that extends radially through the seal and opens into the inner groove to allow the buffer air to flow through the seal into the inner groove.
2. The turbine shroud assembly of claim 1, wherein the seal includes a first seal member and a second seal member coupled to the first seal member for movement therewith, wherein the first seal member has a rectangular cross-sectional shape when viewed in the circumferential direction that defines an inner surface engaged with the shroud wall of the blade track segment, an outer surface spaced apart radially from the inner surface, and two side surfaces spaced apart axially that extend radially between the inner surface and the outer surface, and wherein the second seal member is arranged axially between a first side surface of the first seal member and the carrier segment.
3. The turbine shroud assembly of claim 2, wherein the first seal member is formed to include a side groove that extends axially into a first side surface of the first seal member and circumferentially relative to the axis and the second seal member is arranged in the side groove and engages the carrier segment in the first channel.
4. The turbine shroud assembly of claim 2, wherein the first seal member is formed to include the inner groove that extends radially into the inner surface of the first seal member and circumferentially relative to the axis and the at least one through hole that extends radially through the first seal member and opens into the inner groove to allow the buffer air to flow through the first seal member into the inner groove.
5. The turbine shroud assembly of claim 1, wherein the seal includes an outer seal member, a forward seal member that extends radially inward from the outer seal member, and an aft seal

member that extends radially inward from the outer seal member, the aft seal member spaced apart axially from the forward seal member to define the inner groove therebetween, and the outer seal member, the forward seal member, and aft seal member are free to move relative to each other.

6. The turbine shroud assembly of claim 5, wherein the outer seal member is formed to include the at least one through hole that extends radially through the outer seal member and opens into the inner groove to allow the buffer air to flow through the outer seal member into the inner groove to urge the forward seal member axially forward and the aft seal member axially aft into engagement with the carrier segment.

7. The turbine shroud assembly of claim 1, wherein the seal includes a first seal member and a second seal member coupled to the first seal member for movement therewith, wherein the first seal member has a rectangular cross-sectional shape when viewed in the circumferential direction that defines an inner surface, an outer surface spaced apart radially from the inner surface, and two side surfaces spaced apart axially that extend radially between the inner surface and the outer surface, and wherein the second seal member extends around a portion of the first seal member so that the second seal member engages the carrier segment and the blade track segment.

8. The turbine shroud assembly of claim 7, wherein the first seal member is formed to include the inner groove that extends radially into the inner surface of the first seal member and circumferentially relative to the axis and the at least one through hole that extends radially through the first seal member and opens into the inner groove to allow the buffer air to flow through the first seal member into the inner groove.

9. The turbine shroud assembly of claim 1, wherein the first channel has a rectangular cross-sectional shape viewed circumferentially relative to the axis.

10. The turbine shroud assembly of claim 9, wherein the first channel defines a first end surface, a first support-wall surface that extends radially inward from the first end surface, and a second support-wall surface that extends radially inward from the first end surface at a location spaced apart axially from the first support-wall surface.

11. A turbine shroud assembly for use with a gas turbine engine, the turbine shroud assembly comprising: a carrier segment arranged circumferentially at least partway around an axis, the carrier segment including an outer wall, a first support wall that extends radially inward from the outer wall, and a second support wall that extends radially inward from the outer wall at a location spaced apart axially from the first support wall to define an attachment-receiving space, and the first support wall formed to include a radially-inwardly opening first channel that extends circumferentially relative to the axis and at least one buffer air passageway that extends radially into the forward support wall and opens into the first channel, a blade track segment arranged circumferentially at least partway around the axis to define a portion of a gas path of the gas turbine engine, the blade track segment having a shroud wall that extends circumferentially partway around the axis and an attachment feature that extends radially outward from the shroud wall into the attachment-receiving space formed in the carrier segment, and a buffer air seal assembly arranged in the first channel between the forward support wall of the carrier segment and the shroud wall of the blade track segment to block gases from flowing between the carrier segment and the blade track segment, the buffer air seal assembly including a seal that extends circumferentially relative to the axis and axially across the first channel, wherein the at least one buffer air passageway is configured to discharge buffer air radially inward away from the carrier segment into the first channel to urge the seal radially inward into engagement with the shroud wall of the blade track segment, wherein the seal includes a first seal member and a second seal member coupled to the first seal member for movement therewith, wherein the first seal member has a rectangular cross-sectional shape when viewed in the circumferential direction that defines an inner surface engaged with the shroud wall of the blade track segment, an outer surface spaced apart radially from the inner surface, and two side surfaces spaced apart axially that extend radially between the inner surface and the outer surface, and wherein the second seal member is arranged axially between a

first side surface of the first seal member and the carrier segment, and wherein the first seal member is formed to include an inner groove that extends radially into the inner surface of the first seal member and circumferentially relative to the axis and at least one through hole that extends radially through the first seal member and opens into the inner groove to allow the buffer air to flow through the first seal member into the inner groove.

12. The turbine shroud assembly of claim 11, wherein the first seal member is formed to include a side groove that extends axially into a first side surface of the first seal member and circumferentially relative to the axis and the second seal member is arranged in the side groove and engages the carrier segment in the first channel.

13. A method comprising: providing a carrier segment arranged circumferentially at least partway around an axis, the carrier segment formed to include a radially-inwardly opening first channel and at least one buffer air passageway that extends radially into the carrier segment, providing a blade track segment arranged circumferentially at least partway around the axis, the blade track segment having a shroud wall that extends circumferentially partway around the axis and an attachment feature that extends radially outward from the shroud wall, and the shroud wall formed to include a radially-outwardly opening second channel, providing a buffer air seal assembly including a seal that extends circumferentially relative to the axis and at least one bias member, arranging the at least one bias member of the buffer air seal assembly in the first channel formed in the carrier segment, arranging the seal of the buffer air seal assembly in the first channel formed in the carrier segment so that the bias member is located radially between the carrier segment and the seal, arranging the blade track segment adjacent to the carrier segment so that the seal engages the shroud wall of the blade track segment, and discharging a flow of buffer air through the at least one buffer air passageway radially inward away from the carrier segment toward the seal to urge the seal radially inward into engagement with the blade track segment, wherein the seal is formed to include an inner groove that extends radially inward into the seal and circumferentially relative to the axis and at least one through hole that extends radially through the seal and opens into the inner groove to allow the buffer air to flow through the seal into the inner groove.

14. The method of claim 13, wherein the seal includes a first seal member and a second seal member, and wherein arranging the seal of the buffer air seal assembly in the first channel formed in the carrier segment includes arranging the second seal member in a side groove formed in the first seal member and arranging the first and second seal members in the first channel formed in the carrier segment after arranging the second seal member in the side groove of the first seal member.

15. The method of claim 13, wherein the seal includes an outer seal member, a forward seal member, and an aft seal member, and wherein arranging the seal of the buffer air seal assembly in the first channel formed in the carrier segment includes arranging the outer seal member in the first channel so that the at least one bias member is located radially between the carrier segment and the outer seal member, arranging the forward and aft seal members in the first channel after arranging the outer seal member in the first channel, and spacing the aft seal member apart axially from the forward seal member to define the inner groove therebetween.

16. The method of claim 13, wherein the seal includes a first seal member and a second seal member coupled to the first seal member for movement therewith, wherein the first seal member has a rectangular cross-sectional shape when viewed in the circumferential direction that defines an inner surface, an outer surface spaced apart radially from the inner surface, and two side surfaces spaced apart axially that extend radially between the inner surface and the outer surface, and wherein the second seal member extends around the inner surface of the first seal member.

17. The method of claim 13, further comprising providing at least one retainer and inserting the at least one retainer axially into the carrier segment and through the attachment feature of the blade track segment to couple the blade track segment to the carrier segment.
